

K821 — Ratify Lyra’s Weyl resolution as the RIGHT FRAME (Weyl fermion + ordinary connection = chiral; no self-dual connection, no gravi-weak input, no “universal Born=Bergman” needed — supersedes K820’s endgame). BUT hold the crux HARD: it rests entirely on the Born=Bergman projection genuinely giving a 4D WEYL, not a 4D DIRAC — and the web says the GENERIC dimensional reduction gives DIRAC (vector-like); Weyl requires a nontrivial mechanism. So Lyra’s own fence #1 (Elie’s Weyl-not-Dirac verification) is DECISIVE and could FAIL. If Weyl  $\rightarrow$  parity closes on Born=Bergman alone (clean, no input). If Dirac  $\rightarrow$  back to F636, and the self-dual/gravi-weak input returns.

Keeper | 2026-07-22 | Lyra found the clean finish (Weyl $\rightarrow$ chiral, no input) AND flagged the one verification. Web sharpens WHY that verification is non-trivial and could fail. Ratifying the frame, holding the crux. Beautiful-clean after a retraction-heavy arc = scrutinize hardest.

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Ratify the FRAME (Lyra’s resolution — it’s right, and it removes an input)

The Leg-1/Leg-2 tension collapses onto one fact: a Dirac spinor is vector-like; a Weyl fermion is chiral; Born=Bergman turns the Dirac spinor into a Weyl fermion. - The F636 “vector-like” wall was the UN-projected DIRAC spinor (both chiralities, same rep  $\rightarrow$  gauge hits both). - If Born=Bergman projects to WEYL (one chirality, the holomorphic/positive-energy sector), then an **\*\*ORDINARY SU(2)<sub>L</sub> connection is already chiral\*\*** — because a Weyl fermion has NO opposite-chirality partner for the connection to make vector-like. This is literally how the SM works: SU(2)<sub>L</sub> is an ordinary gauge field, chiral only because it couples to left-handed Weyl doublets while the right-handed fields are singlets. - So “is Born=Bergman universal / is the connection self-dual?” (K820’s endgame) is SUPERSEDED: we don’t need it. The chirality lives in the FERMIONS being Weyl, not the connection being self-dual. No gravi-weak input. The self-dual route (Elie 4780) is a real but UNNEEDED alternative. This is a strengthening (fewer inputs), IF the fermion is genuinely Weyl. - Not a reversal of Lyra’s F639 brake: there she flagged chiral-spectrum  $\nRightarrow$  chiral-coupling (true for a DIRAC fermion); for a WEYL fermion it DOES imply it. Flag  $\rightarrow$  resolve, not flip-flop.

★★ The crux — and the web says it’s genuinely non-trivial (could FAIL)

The ENTIRE resolution rests on one claim: the Born=Bergman projection gives a 4D WEYL fermion, NOT a 4D DIRAC. Lyra’s own fence #1. The web makes this decisive, not a formality: - Web: “a higher-dim Weyl fermion decomposes into a pair of L and R 4d chiralities = a 4d DIRAC fermion.” The GENERIC dimensional reduction gives DIRAC (vector-like) — this IS the Witten no-go / the F636 problem. - Web: “chiral spectrum ... where L or R modes are projected out, requires NONTRIVIAL BACKGROUNDS or SINGULAR POINTS.” Getting WEYL requires a special mechanism, not automatic. - Dimension check (the sharp worry): Lyra’s  $8 = 4_{+1/2} \oplus 4_{-1/2}$ . A 4D Dirac spinor is 4-complex-dim; a 4D Weyl is 2. If the  $4_{+1/2}$  half is a 4D DIRAC (4-dim), keeping it is VECTOR-LIKE — F636 again. If it genuinely splits into 4D Weyls, it’s chiral. This is NOT obvious from the dimension count — it’s exactly what must be verified. - So the decisive question: does the positive-energy/holomorphic ( $c=+1/2$ ) projection give ONE 4D spacetime chirality (Weyl  $\rightarrow$  chiral, parity closes) or a full 4D Dirac (both chiralities  $\rightarrow$  vector-like  $\rightarrow$  F636, and the self-dual input returns)?

The two routes and what the verification decides

- Route B (Lyra, clean): the FERMION is Weyl (Born=Bergman) + ordinary connection  $\rightarrow$  chiral. No input. Requires the Weyl-not-Dirac verification to succeed.
- Route A (Elie 4780, fallback): the CONNECTION is self-dual/holomorphic  $\rightarrow$  chiral (even with a Dirac fermion). Needs the gravi-weak input.

- The Weyl-not-Dirac verification decides which: Weyl  $\rightarrow$  Route B, clean, no input, parity closes on Born=Bergman alone. Dirac  $\rightarrow$  Route A, parity derived-CONDITIONAL on the self-dual/gravi-weak input.

### The finish (honest)

- This is a CANDIDATE, held HARD. The frame is right (Weyl $\rightarrow$ chiral); the crux (Weyl-not-Dirac) is decisive and — per the web — could genuinely go DIRAC (the generic answer). Do NOT bank “parity closes on Born=Bergman alone” until Elie’s explicit 4D reduction confirms WEYL, not Dirac. Beautiful-clean earns extra suspicion after this arc’s retractions.
- The verification IS the finish: Elie’s explicit  $SO(5,2)\rightarrow SO(3,1)$  reduction of the  $c=+1/2$  holomorphic sector — is it a 4D Weyl (2-dim chiral) or a 4D Dirac (4-dim vector-like)? That single check ends the thread, one way or the other.
- Stays open regardless (separate legs, Lyra): which  $SU(2)$  is electroweak ( $g=7$  orientation / why  $SU(2)_L$ ); hypercharge origin (center, K806); the doublet/singlet ADDRESS assignment (= the flavor K-type problem). The CORE parity violation is what this closes (or not); the representation structure is separate.
- Survivors bank: chirality MECHANISM (chamber theorem), T2520, T2521, split, CP-free.

— Keeper K821, 2026-07-22. Ratify Lyra’s FRAME (Weyl fermion + ordinary connection = chiral; no self-dual, no universal-Born=Bergman, no gravi-weak input — supersedes K820). BUT the crux is DECISIVE and could FAIL: it rests entirely on Born=Bergman giving a 4D WEYL not DIRAC, and the web says the GENERIC reduction gives DIRAC (vector-like, = F636); Weyl needs a nontrivial mechanism. Dimension worry: is  $4_{\{+1/2\}}$  a 4D Dirac (4-dim, vector-like) or Weyl? Elie’s explicit reduction decides: Weyl  $\rightarrow$  parity closes on Born=Bergman alone (clean, no input); Dirac  $\rightarrow$  self-dual/gravi-weak input returns (Route A). CANDIDATE held HARD; don’t bank until Weyl confirmed. Open legs: which  $SU(2)$ , hypercharge (K806), doublet/singlet addresses. See [[Keeper\_K820\_ratify\_spectrum\_is\_not\_coupling\_two\_legs\_separate\_coupling\_leg\_is\_holomorphic\_connection\_quark\_mass\_a\_07-22]], Lyra (Weyl resolution + fence #1), Elie 4780 (self-dual route), F636 (vector-like), Witten no-go / web (generic reduction = Dirac).